

$C^1(0, 1 | \mathcal{H}_0)/n + C^2(0, 1 | \mathcal{H}_0)/n^2$ have a non-negative argument; for $\kappa_0 = 0.05$ we have that $C^1(0, 1 | \mathcal{H}_0) = -199.83$. Note that under the null hypothesis, the Laplace approximations are accurate sooner than under the alternative hypothesis, because the priors are already concentrated at zero. Under the null hypothesis the general observation remains true that for reasonable sample sizes the expected peri-null Bayes factor is far from the limiting value.

Unlike the peri-null Bayes factor, the (default) point-null Bayes factor is consistent. Fig. 3 shows the simulated sampling distribution of the point-null and peri-null Bayes factors in blue and black respectively. As before the 97.5% quantile (top dotted curve), the average (solid curve), and the 2.5% quantile (bottom dotted curve) are depicted as well.

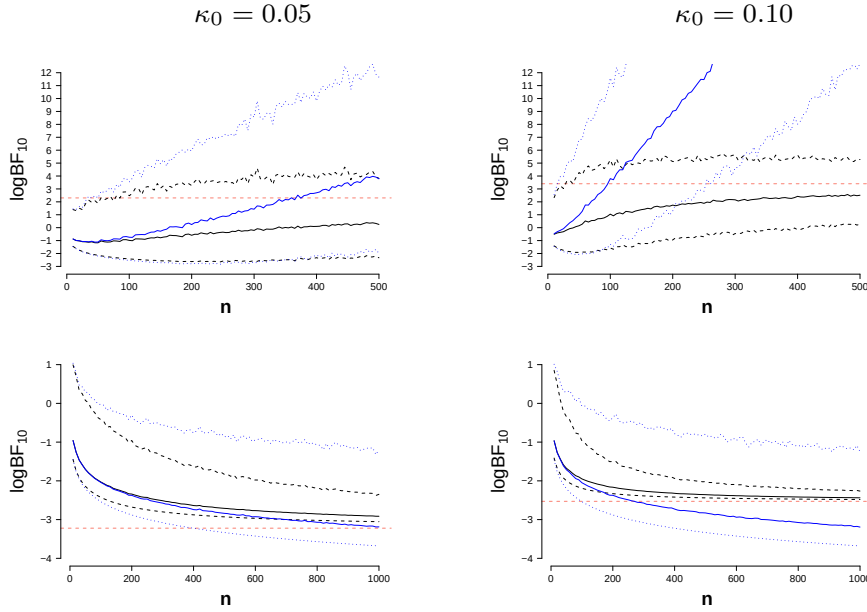


Fig. 3: (Default) point-null Bayes factor t -tests (depicted in blue) are consistent under both the alternative and null, e.g., top and bottom row respectively, as opposed to peri-null Bayes factors (depicted in black). Note that the peri-null and the default point-null Bayes factors behave similarly when n is small. The domain where the two types of Bayes factors behave similarly is smaller when the peri-null is less concentrated, e.g., compare the right to the left column.

The top left subplot of Fig. 3 shows that under $\mu = 0.167$ and $\sigma = 1$ the point-null and peri-null Bayes factor behave similarly up to $n = 30$. Furthermore, the average point-null log Bayes factor crosses the peri-null upper bound of $\log(10)$ at around $n = 380$, whereas the peri-null Bayes factor remains bounded even in the limit, and is therefore inconsistent. The top right subplot shows, under $\mu = 0.348$ and $\sigma = 1$, that the discrepancy between the point-

null and peri-null Bayes factor becomes apparent sooner when the peri-null prior is less concentrated, i.e., $\kappa_0 = 0.10$ instead of $\kappa_0 = 0.05$. Also note that under these alternatives, the logarithm of the point-null Bayes factor grows linearly (e.g., [Bahadur and Bickel, 2009](#); [Johnson and Rossell, 2010](#)). Hence, the point-null Bayes factor has a larger power to detect an effect than that afforded by the peri-null Bayes factor.

The bottom row of Fig. 3 paints a similar picture; under the null the point-null Bayes factor accumulates evidence for the null hypothesis without bound as n grows. For $\kappa_0 = 0.05$ the behavior of the peri-null and the point-null Bayes factor is similar up to $n = 200$ and it takes about $n = 1,000$ samples before the average point-null log Bayes factor crosses the peri-null lower bound of -3.22 . For $\kappa_1 = 0.10$ only $n = 270$ samples are needed before the log Bayes factor for the point-null hypothesis crosses the peri-null lower bound of -2.53 .

5 Towards Consistent Peri-Null Bayes Factors

There are at least three methods to adjust the peri-null Bayes factor in order to avoid inconsistency. The first method changes both the point-null hypothesis $\mathcal{H}_0$ and the alternative hypothesis $\mathcal{H}_1$. Specifically, one may define the hypotheses under test to be non-overlapping (e.g., [Berger and Delampady, 1987](#); [Chandramouli and Shiffrin, 2019](#); [Rousseau, 2007](#)). The resulting procedure is usually known as an ‘interval-null hypothesis’, where the interval-null is defined as a (renormalized) slice of the prior distribution for the test-relevant parameter under an alternative hypothesis (e.g., [Morey and Rouder, 2011](#)). For instance, in the case of a t -test an *encompassing* hypothesis $\mathcal{H}_e$ may assign effect size δ a Cauchy distribution with location parameter 0 and scale κ_e ; from this encompassing hypothesis one may construct two rival hypotheses by restricting the Cauchy prior to particular intervals: the interval-null hypothesis truncates the encompassing Cauchy distribution to an interval centered on $\delta = 0$: $\delta \sim \text{Cauchy}(0, \kappa_e)I(-a, a)$, whereas the interval-alternative hypothesis is the conjunction of the remaining two intervals, $\delta \sim \text{Cauchy}(0, \kappa_e)I(-\infty, -a)$ and $\delta \sim \text{Cauchy}(0, \kappa_e)I(a, \infty)$. As a consequence of Theorem 1, or Theorem 1 (ii) of [Dawid \(2011\)](#), the resulting peri-null Bayes factor is then consistent in accordance to subjective interval belief; for all data-governing parameters δ in the interior of the interval-null, $\lim_{n \rightarrow \infty} \text{BF}_{10} = 0$, and for δ in the interior of the sliced out alternative $\lim_{n \rightarrow \infty} \text{BF}_{01} = 0$.⁵ In particular, when $a = 1$ and the data-governing $\delta = 0.7$, then this Bayes factor will eventually show unbounded evidence for the interval-null.

One disadvantage of this method is the need to specify the width of the interval ([Jeffreys, 1961](#), p. 367). This disadvantage can be mitigated by reporting a range of non-overlapping interval-null Bayes factors as a function of a ; the researcher can then draw their own conclusion. The resulting range of interval-null Bayes factors also respects the uncertainty about the proper

⁵ For consistency to hold the standard condition is assumed that the interval-null or sliced up prior assigns positive mass to a neighborhood of δ in the respective intervals.

specification of the interval-null hypothesis and thereby avoids a false sense of precision.⁶ A second disadvantage of the non-overlapping interval-null method is that the prior distributions for the rival interval hypotheses are of an unusual shape – a continuous distribution up to the point of truncation, where the prior mass abruptly drops to zero. It is debatable whether such artificial forms would ever result from an elicitation effort. A third disadvantage is that it seems somewhat circuitous to parry the critique “the null hypothesis is never true exactly” by adjusting both the null hypothesis *and* the alternative hypothesis.

The second method to specify a (partially) consistent peri-null Bayes factor is to change the point-null hypothesis to a peri-null hypothesis by supplementing rather than supplanting the spike with a distribution (Morey and Rouder, 2011). In other words, the point-null hypothesis is upgraded to include a narrow distribution around the spike. This mixture distribution is generally known as a ‘spike-and-slab’ prior, but here the slab represents the peri-null hypothesis and is relatively peaked. This mixture model $\mathcal{H}_{0'}$ may be called a ‘hybrid null hypothesis’ (Morey and Rouder, 2011), a ‘mixture null hypothesis’, or a ‘peri-point null hypothesis’. Thus, $\mathcal{H}_{0'} = \xi\mathcal{H}_0 + (1 - \xi)\mathcal{H}_{\tilde{0}}$, with $\xi \in (0, 1)$ the mixture weights and, say, $\xi = \frac{1}{2}$. Because $\xi > 0$, the Bayes factor comparing $\mathcal{H}_{0'}$ to $\mathcal{H}_1$ will be consistent when the data come from $\mathcal{H}_0$; and because $\mathcal{H}_{0'}$ also has mass away from the point under test, the presence of a tiny true non-zero effect will not lead to the certain rejection of the null hypothesis as n grows large. The data determine which of the two peri-point components receives the most weight. As before, for modest sample sizes and small κ_0 , the distinction between point-null, peri-null, and peri-point null is immaterial. The main drawback of the peri-point null hypothesis is that it is consistent only when the data come from $\mathcal{H}_0$; when the data come from $\mathcal{H}_1$ or $\mathcal{H}_{\tilde{0}}$, the Bayes factor remains bounded as before (i.e., Eq. 8).

The third method is to define a peri-null hypothesis whose width κ_0 slowly decreases with sample size (i.e., a ‘shrinking peri-null hypothesis’). For the t -test, one can take $\kappa_0 = c\sigma/\sqrt{n}$ for some constant $c > 0$ as proposed by Berger and Delampady (1987), see also Rousseau (2007), except that their proposal involves a test of non-overlapping hypotheses. More generally, consistency follows by adapting Theorem 1, which depends on the Laplace approximation that becomes invalid if κ_0 shrinks too quickly. The representation Eq. (3) shows that this consistency fix is equivalent to keeping the peri-null correction Bayes factor $\text{BF}_{\tilde{0}0}$ close to one regardless of the data. Note that this is attainable as $\kappa_0 \rightarrow 0$ the peri-null and point-null become indistinguishable. In other words, the consistency of such a shrinking peri-null Bayes factor is essentially driven by the asymptotic behavior of the point-null Bayes factor; arguably, one might as well employ this point-null Bayes factor to begin with. One drawback of the shrinking peri-null hypothesis is that it is incoherent,

⁶ We thank the first anonymous reviewer for suggesting this procedure to circumvent a definite choice for a .